From multiscale quantum mechanics to Newton's initial-value problem

Result, 2026-10-27. For a smooth nonnegative potential on the line with bounded second and higher derivatives, we construct Euclidean 0+1 quantum mechanics at every supplied $h > 0$ using time refinement and a normalized density at every dyadic scale. The reference fluctuations are Gaussian Brownian bridges, and the full interaction and large-field configurations are retained. Real-time evolution is the unitary group of the reconstructed self-adjoint Hamiltonian.

For every initial (q_0, p_0), means of finite quantum time-slice products are a Cauchy family as both h and the time mesh tend to zero. An explicit $O(h + |\pi|)$ estimate controls both scales simultaneously for the means, while the packet widths remain $O(\sqrt{h})$. Their limit solves $\dot{q} = p/m$, $\dot{p} = -V'(q)$; it is the unique solution. An explicit bound compares any two positive values of h. The proof does not assume a classical trajectory in advance. Its stability hypothesis and Gronwall estimate are the same ingredients used by Cauchy–Lipschitz, so the construction is a different route, not a weakening of that theorem. For smooth confining potentials with locally Lipschitz force, including the positive quartic, a cutoff and energy conservation construct the classical trajectory from quantum families with cutoff potentials. This argument does not prove the joint approximation estimate for the uncut quartic quantum evolution.

For a didactic entry point, see Newton, quantum mechanics and Dimock–Balaban: it traces the steps to specific source passages and introduces existence, uniqueness, continuation, magnetic forces and a separate hard-wall example. The theorem of this note remains the smooth scalar-potential construction stated above.

This note is a new assembly of standard Feynman–Kac, conditional expectation and unitary-conjugation ingredients; no priority claim is made. The normalized-density architecture is inspired by Dimock–Balaban, not an import of their four-dimensional Yang–Mills estimates. The construction assumes a positive action parameter h, and does not prove it necessary. A separate tangent-groupoid formulation follows only after this theorem.

Targeted advisor check (GPT-6.1 Sol, 2026-10-27). The finite-step identities, constants, joint Cauchy estimate, domain argument, quartic energy barrier and cusp counterexample were checked in writing. This is not an independent proof audit of the imported Wiener, Feynman–Kac, self-adjointness or spectral theorems. The centered $O(h)$ improvement below follows a subsequent user-suggested calculation and is not covered by that earlier check.

1. Hypotheses and the quantum theory to be constructed

Let $m > 0$ and let $V : \mathbb{R} \to \mathbb{R}$ be C^∞, $V \geq 0$, with all derivatives of order at least two bounded. Set

$$L = \sup_x |V''(x)| < \infty. \tag{1}$$

Thus V' is globally Lipschitz with constant L and V grows at most quadratically. A nonlinear confining example is

$$V(x) = \tfrac{1}{2} m\omega_0^2 x^2 + \lambda[1 - \cos(x/\ell)], \qquad \omega_0, \lambda, \ell > 0. \tag{2}$$

Its curvature is bounded, while the force is not linear.

For each $h > 0$, set

$$H_h = -\frac{h^2}{2m}\frac{d^2}{dx^2} + V(x) \tag{3}$$

on $L^2(\mathbb{R})$. Its Friedrichs realization is nonnegative and self-adjoint; for this smooth bounded-below class the restriction to $C_c^\infty(\mathbb{R})$ is a core. Put

$$U_h(t) = e^{-itH_h/h}. \tag{4}$$

The spectral theorem makes U_h a strongly continuous unitary group. No classical ODE solution is used to define it. If $V(x) \to \infty$ at infinity, compactness of the form embedding and positivity improvement additionally give the unique vacuum and positive spectral gap as in the preceding construction. They are not required for the initial-value theorem below.

2. Finite time weights and arbitrary refinement

Fix endpoints x, y, duration $T > 0$, and a partition $\pi = (0 = t_0 < \cdots < t_N = T)$. Write $\delta_j = t_{j+1} - t_j$ and

$$S_\pi^E(q) = \sum_j \left\{ \frac{m(q_{j+1} - q_j)^2}{2\delta_j} + \frac{\delta_j}{2}[V(q_j) + V(q_{j+1})] \right\}. \tag{5}$$

The normalized finite-lattice measure has endpoints $q_0 = x, q_N = y$ and density proportional to $e^{-S_\pi^E/h}$ in the internal coordinates.

Let B be the Brownian bridge from x to y with variance parameter h/m and mean $b(t) = x + (y - x)t/T$. Its covariance is

$$\mathrm{Cov}[B(s), B(t)] = \frac{h}{m}\left(\min(s,t) - \frac{st}{T}\right). \tag{6}$$

With $p_h(t; u, v) = \sqrt{m/(2\pi h t)}\exp[-m(v-u)^2/(2ht)]$, the normalized kinetic factor $\prod_j p_h(\delta_j; q_j, q_{j+1})/p_h(T; x, y)$ is exactly the joint density of $B(t_j)$. Define

$$W_\pi(B) = \sum_j \frac{\delta_j}{2}[V(B(t_j)) + V(B(t_{j+1}))], \qquad Z_\pi = \mathbb{E}_0 e^{-W_\pi(B)/h}. \tag{7}$$

The interpolated lattice measure is the pushforward of $Z_\pi^{-1} e^{-W_\pi/h} d\mathbb{P}_0$ by polygonal interpolation.

Brownian continuity gives uniform convergence of this interpolation to B, and $W_\pi \to \int_0^T V(B(t))dt$ almost surely, for every sequence of partitions with $|\pi| \to 0$. The weights lie in $[0, 1]$. Consequently dominated convergence constructs the limit

$$d\nu_h = Z_h^{-1} e^{-h^{-1}\int_0^T V(B(t))dt} d\mathbb{P}_0, \qquad Z_h = \mathbb{E}_0 e^{-h^{-1}\int_0^T V(B(t))dt} > 0. \tag{8}$$

It proves weak convergence of the polygonal measures on $C([0, T])$ and $L^1(\mathbb{P}_0)$ convergence of their **lifted densities**. The polygonal measures themselves are not asserted to converge in total variation to the non-polygonal path measure.

The normalization has an explicit finite-partition lower bound. Since $V'' \le L$, Taylor's inequality gives $\mathbb{E}V(B(t)) \le V(b(t)) + \frac{L}{2}\operatorname{Var} B(t)$. Writing $M_{xy} = \max_{z\in[x,y]} V(z)$, using $\operatorname{Var} B(t) \le hT/(4m)$ and Jensen's inequality yields

$$Z_\pi \ge \exp\left[-\frac{TM_{xy}}{h} - \frac{LT^2}{8m}\right] \quad \text{for every partition.} \tag{9}$$

Thus normalization and large-field paths are included at every mesh. The bound is finite for each $h > 0$ but is not uniform as $h \to 0$.

3. The exact all-scale retained density

For the dyadic partition let $\mathcal{F}_n$ be the sigma-algebra of the $2^n + 1$ vertex values, and let

$$\rho = \frac{d\nu_h}{d\mathbb{P}_0}, \qquad \rho_n = \mathbb{E}_0[\rho \mid \mathcal{F}_n]. \tag{10}$$

These are densities relative to the **normalized free bridge**. Conditional expectation gives, exactly,

$$\rho_n = \mathbb{E}_0[\rho_{n+1} \mid \mathcal{F}_n], \quad \mathbb{E}_0\rho_n = 1, \quad 0 < \rho_n \le Z_h^{-1}. \tag{11}$$

At level $n+1$, conditional on level n, the reference midpoint of a cell of length $\delta = T/2^n$ has mean $(a+b)/2$ and variance $h\delta/(4m)$; reference midpoints in different cells are independent. The *interacting* conditional law is the product of these Gaussian reference laws multiplied by ρ_{n+1}/ρ_n. The latter density must be retained; the interacting midpoint variables are not assumed independent.

Dyadic evaluations generate the Borel sigma-algebra of continuous path space. Hence the bounded martingale theorem gives

$$\rho_n \longrightarrow \rho \quad \text{in } L^1(\mathbb{P}_0) \text{ and almost surely.} \tag{12}$$

Equations (11)–(12) are an exact normalized blocking tower at all scales. They use an already constructed Wiener measure as reference; they do not claim a lattice-only construction or a four-dimensional polymer estimate. In particular the Z_h^{-1} bound does not give control uniform in $h \downarrow 0$.

The distinction between a finite lattice step and the complete effective density can be seen explicitly. Before any blocking, integrate the middle vertex z of two cells of length s, with retained endpoints a, b and $\mu = (a+b)/2$. Its conditional potential is

$$F_\mu(z) = \frac{m}{s}(z-\mu)^2 + sV(z). \tag{13}$$

For $2m/s - sL > 0$, put $\kappa_s = 2m/s - sL$. Then $F_\mu'' \ge \kappa_s$; there is a unique minimizer z_* and the **normalized** conditional law satisfies

$$\mathbb{P}(|z - z_*| \ge r \mid a, b) \le 2\exp[-\kappa_s r^2/(2h)]. \tag{14}$$

Indeed, integration by parts gives $\mathbb{E}(z-z_*)^{2n} \le (2n-1)!!(h/\kappa_s)^n$; expand the hyperbolic cosine and optimize the exponential Markov bound. Conditioned on all retained vertices, the eliminated vertices are independent for this original nearest-neighbour lattice. After a blocking step the generated action must be kept: (13)–(14) cannot be iterated as though the original potential were unchanged.

This is the precise $0+1$ version of the source lesson: retain the entire normalized effective density, not merely its background minimizer or an unnormalized large-field action cost. Compare Dimock I, sections 2.1, 2.4 and 4.5, and Dimock II, introduction and sections 3.13–3.14 (selected passages read there). The latter papers treat a three-dimensional scalar theory, not (3).

4. From the Euclidean limit to real-time quantum mechanics

The Feynman–Kac theorem identifies the kernel of $e^{-TH_h/h}$ as $K_h(T;x,y) = p_h(T;x,y)Z_h(x,y)$; (8) is the corresponding **normalized bridge**, not the kernel itself. It also makes the semigroup reflection positive in the ground-state construction when a normalizable ground state exists. Real time is defined by the self-adjoint operator through (4), not by treating the Euclidean probability law as a real-time probability law.

For finite real-time steps use the kick–drift product

$$T_{h,s} = D_{h,s}K_{h,s}, \quad K_{h,s} = e^{-isV(Q)/h}, \quad D_{h,s} = e^{-isP^2/(2mh)}, \quad P = -ih\,\partial_x. \tag{15}$$

The factors are unitary. Its products converge strongly to $U_h(T)$ under arbitrary positive partitions at each fixed $h > 0$. Here is the needed arbitrary-partition argument.

Write $c_h = h^2/(2m)$, and take $f \in C_c^\infty$. Integration by parts gives

$$\|H_h f\|^2 = c_h^2\|f''\|^2 + \|Vf\|^2 + 2c_h \int V|f'|^2 - c_h \int V''|f|^2 \geq c_h^2\|f''\|^2 + \|Vf\|^2 - c_h L\|f\|^2. \quad (15a)$$

Since C_c^∞ is an operator core, closure identifies $D(H_h) = D(P^2) \cap D(V)$ and controls the two component graph norms. For f in this domain, $(T_{h,s} - I)f/s \to -iH_h f/h$ strongly, and these difference quotients are uniformly bounded from the H_h graph domain to L^2. The orbit $\{U_h(t)f : 0 \leq t \leq T\}$ is graph-norm compact. Thus the local first-order error is uniform on the orbit, and a telescoping sum over the cells proves the limit; density and unitarity extend it to all L^2 vectors. This is the argument used for the symmetric product in Rivero 1998 test, Theorem 2(b), now written for kick–drift. It is a strong operator statement, not a finite complex measure on path space.

The Euclidean stationary equation is $m\ddot{q} = +V'(q)$, whereas real-time Newton motion satisfies $m\ddot{q} = -V'(q)$. Euclidean concentration cannot be used as a Newton trajectory by silently changing that sign.

5. A joint quantum and time-mesh construction of Newton motion

Fix $\omega > 0$, initial (q_0, p_0), and the coherent packet

$$\psi_h(x) = \left(\frac{m\omega}{\pi h}\right)^{1/4} \exp\left[-\frac{m\omega(x-q_0)^2}{2h} + \frac{ip_0(x-q_0)}{h}\right]. \quad (16)$$

Its position and momentum means are q_0, p_0, with standard deviations $\sqrt{h/(2m\omega)}$ and $\sqrt{hm\omega/2}$.

For a partition π, apply the unitary kick and drift (15) successively. Let q_j, p_j be the resulting means at t_j, and put

$$W_j = \mathrm{sd}(Q)_j + \frac{\mathrm{sd}(P)_j}{m\omega}, \quad C = \sqrt{\frac{2}{m\omega}}, \quad \beta = \omega + \frac{L}{m\omega}, \quad a = \max\left(\omega, \frac{L}{m\omega}\right). \quad (17)$$

No Gaussian assumption is made about the state after the first step.

The exact conjugation identities are

$$K_{h,s}^* Q K_{h,s} = Q, \quad K_{h,s}^* P K_{h,s} = P - sV'(Q), \quad D_{h,s}^* Q D_{h,s} = Q + sP/m, \quad D_{h,s}^* P D_{h,s} = P. \quad (18)$$

Both finite factors preserve Schwartz space under the stated derivative bounds, so (18) is valid on each propagated Gaussian. For the kick, the triangle inequality in L^2 and $\mathrm{sd}[V'(Q)] \leq L\,\mathrm{sd}(Q)$ give $\mathrm{sd}(P)_{\mathrm{after}} \leq \mathrm{sd}(P)_{\mathrm{before}} + sL\,\mathrm{sd}(Q)$. For the drift the analogous bound adds $s\,\mathrm{sd}(P)/m$ to the position deviation. Therefore

$$W_{j+1} \leq (1 + s\omega)\left(1 + \frac{sL}{m\omega}\right) W_j, \qquad W_j \leq C\sqrt{h}\, e^{\beta T}. \quad (19)$$

These are finite-step statements for the full quantum state.

The means obey

$$p_{j+1} = p_j - s_j \langle V'(Q)\rangle_j, \qquad q_{j+1} = q_j + \frac{s_j p_{j+1}}{m}. \quad (20)$$

Fix $h_0 > 0$, restrict $0 < h \leq h_0$, and use the phase-space norm $\|(q,p)\|_\omega = |q| + |p|/(m\omega)$. Define

$$\begin{gathered} b = \frac{|V'(0)| + LC\sqrt{h_0}e^{\beta T}}{m\omega}, \\ R_T = e^{\beta T}(\|(q_0,p_0)\|_\omega + Tb), \qquad F_T = |V'(0)| + L(R_T + C\sqrt{h_0}e^{\beta T}), \\ C_\delta = \frac{F_T + LR_T}{m}, \qquad C_h = \frac{\|V'''\|_\infty C^2 e^{2\beta T}}{2m\omega}, \qquad G_T = \frac{e^{aT} - 1}{a}. \end{gathered} \quad (21)$$

Equation (19) and (20) imply recursively that every vertex mean has $\|(q_j, p_j)\|_\omega \le R_T$ and $|\langle V'(Q)\rangle_j| \le F_T$, independently of h and π. For example, the recurrence for the norm is bounded by $R_{j+1} \le e^{\beta s_j} R_j + s_j b e^{\omega s_j}$.

Linearly interpolate the mean vertices, writing $z_{h,\pi}(t) = (q_{h,\pi}(t), p_{h,\pi}(t))$. Put $\mathcal{F}(q,p) = (p/m, -V'(q))$. On every cell the two components of the residual are bounded by

$$|\dot{q}_{h,\pi} - p_{h,\pi}/m| \le |\pi| F_T/m, \quad |\dot{p}_{h,\pi} + V'(q_{h,\pi})| \le \tfrac{1}{2}\|V'''\|_\infty C^2 h\, e^{2\beta T} + L|\pi|\,\omega R_T. \tag{22}$$

Consequently

$$\|\dot{z}_{h,\pi} - \mathcal{F}(z_{h,\pi})\|_\omega \le C_\delta |\pi| + C_h h. \tag{23}$$

The first residual is the difference between the interpolated momentum and its post-kick value; the second is the sum of the packet's force bias and the motion of the interpolated position. For the bias, Taylor expand V' around the packet mean $q_j = \langle Q\rangle_j$. Its linear term has zero expectation, so

$$|\langle V'(Q)\rangle_j - V'(q_j)| \le \tfrac{1}{2}\|V'''\|_\infty \operatorname{Var}_j(Q) \le \tfrac{1}{2}\|V'''\|_\infty C^2 h\, e^{2\beta T}. \tag{23a}$$

The bounded third derivative is part of the standing hypotheses. For quadratic potentials $V''' = 0$ and the force-bias term vanishes exactly. This improves the mean error; (19) still gives widths of order $\sqrt{h}$.

The vector field $\mathcal{F}$ has Lipschitz constant at most a in this norm. Integrating the residuals of two products with common initial data, and applying Gronwall, gives

$$\sup_{t\le T} \|z_{h,\pi}(t) - z_{k,\sigma}(t)\|_\omega \le G_T[C_\delta(|\pi| + |\sigma|) + C_h(h+k)]\,. \tag{24}$$

Thus the means are Cauchy in $C([0,T],\mathbb{R}^2)$ as the two scales h and $|\pi|$ tend to zero **in any joint order**. Let $z = (q,p)$ be their limit. Equation (23) passes to the integral equation

$$q(t) = q_0 + \int_0^t \frac{p(s)}{m}\, ds, \qquad p(t) = p_0 - \int_0^t V'(q(s))\, ds. \tag{25}$$

It is continuously differentiable and solves Newton's initial-value problem. Uniqueness follows from the same Gronwall inequality with zero residual. The construction extends to negative time using inverse unitary steps, or by replacing p by $-p$ and reversing time.

Letting $(k, |\sigma|) \to (0,0)$ in (24) yields the explicit joint bound

$$\sup_{t\le T} \|z_{h,\pi}(t) - z(t)\|_\omega \le G_T\big(C_\delta|\pi| + C_h h\big). \tag{26}$$

The constants are independent of the number of cells and of h in $(0, h_0]$. Equation (26) is an observable-level statement about the packet means; it is not an operator-norm estimate on real-time propagators and does not assert a real-time positive path measure.

Finally fix $h > 0$. At a fixed t, choose partitions containing t. Section 4 gives strong L^2 convergence of the product state to $U_h(t)\psi_h$. Equations (19)–(21) bound its position and momentum second moments uniformly. Strong L^2 convergence gives convergence of bounded position and momentum cutoffs, and the uniform second moments make the first-moment tails at most $O(R^{-1})$. The first moments therefore converge to those of $U_h(t)\psi_h$. For a general refining partition, use the product truncated at the last vertex before t; the same graph-norm telescoping argument is uniform on compact time intervals. The bounds on vertex momenta and forces make the interpolated means equi-Lipschitz. Pointwise identification therefore upgrades to uniform convergence on $[0,T]$. Passing to the limit in (26) gives

$$\sup_{t\le T} \|(\langle Q\rangle_t, \langle P\rangle_t) - z(t)\|_\omega \le G_T C_h h. \tag{27}$$

This uses no differentiation of an unbounded quantum observable. It also proves the claimed consistency of the two limiting orders for these means. The construction assumes a semiclassical family $h \downarrow 0$; a single fixed-positive-h theory does not have simultaneously sharp position and momentum.

The Cauchy estimate and its domain argument are the present assembly. The broader relation between coherent quantum packets and classical orbits is established prior work, including Hepp, 1974 (introduction and theorem statements); no claim of literature novelty is made for this semiclassical mechanism.

6. Quartic extension and the limits of the implication

Let $V(q) = \lambda q^4/4$ with $\lambda > 0$. Its Hessian is unbounded, so (26) does not apply globally with a single L. For fixed initial data choose a smooth nonnegative V_R with bounded derivatives of order at least two, agreeing with V on $[-R, R]$. Apply the construction to V_R. As long as its trajectory is inside that interval,

$$E = \frac{p^2}{2m} + \frac{\lambda q^4}{4} = \frac{p_0^2}{2m} + \frac{\lambda q_0^4}{4} \tag{28}$$

by the chain rule. Choosing $R > (4E/\lambda)^{1/4}$ rules out a first exit, so the constructed trajectory solves the original quartic equation for all finite times. Local Lipschitz uniqueness identifies different choices of cutoff. The same argument applies to smooth confining potentials whose derivative is locally Lipschitz, with an appropriate compact energy sublevel. The cutoff quantum means are not claimed here to be those of the uncut quartic Hamiltonian.

Quantum existence does **not** alone imply classical uniqueness. The bounded C^1 potential in the cusp note, with $V(q) = -\frac{2}{3}|q|^{3/2}$ near zero, has a bounded-perturbation self-adjoint quantum Hamiltonian; adding a constant makes $V \geq 0$ without changing the force. Yet, while remaining in that neighbourhood, Newton's equation with $m = 1$ admits both rest and

$$q(t) = \begin{cases} 0, & t \leq t_0, \\ (t - t_0)^4/144, & t \geq t_0, \end{cases} \tag{29}$$

for any waiting time $t_0 \geq 0$. Thus extra force regularity, or some independent selection rule, is necessary for the uniqueness conclusion.

7. Comparison with Cauchy–Lipschitz

The direct theorem works with the integral operator on phase-space paths; Lipschitz continuity gives a contraction locally, then global continuation under the global hypothesis. The earlier polygon lesson gives a comparable refinement/Cauchy proof.

The route above first constructs a quantum family and then proves that its means form a Cauchy family as $h \downarrow 0$. Its quantitative inputs are normalized multiscale Euclidean control, self-adjoint real-time evolution, a packet variance estimate, and the **same** Lipschitz/Gronwall stability that makes Cauchy–Lipschitz work. It gives the explicit approximation (26) and places the classical path beside quantum dynamics, but it invokes substantially more machinery and does not establish an ODE theorem beyond Cauchy–Lipschitz.

8. Consequence for STATE

The user-directed first document gives a complete scoped existence and classical-limit construction. The subsequent tangent-groupoid formulation places its observables and dynamics in a continuous field, keeping the deformation parameter, time mesh and coarse-graining distinct while relating their observable-state limits along chosen joint schedules.